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CHAPTER I

Capital Markets and Investment Performance

Returns, risk measurement, the equity premium, and historical capital markets data.

Overview

Securities are nothing more than promises of future payment, distributed through intermediaries like investment banks. These can be traded on exchanges such as the NYSE, AMEX, and NASDAQ.

Securities vary widely in complexity, from straightforward bonds to sophisticated structured notes. The fundamental characteristic shared across all securities is that they represent economic claims against future benefits, always incorporating a time dimension.

Key Concept

All securities, no matter how complex, share one essential feature: they are **economic claims against future benefits**, and therefore always incorporate a time dimension.

I. Finance from the Investor's Perspective

This module shifts focus from corporate decision-making to the investor's perspective. We examine how shareholders and bondholders who hold portfolios — collections of different securities — make financial decisions. The central question addressed is:

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II. Why Investors Invest

Primary motivations for purchasing securities include savings for future needs and wealth accumulation. Some investors accept high-risk ventures, such as lottery tickets, for potential significant payoffs. Additional motivations include charitable objectives, though these generate non-economic dividends and prove difficult to evaluate systematically.

III. Definition of Rates of Return

Return represents the growth in wealth resulting from an investment, expressed as a percentage for comparability. The basic formula is:

SIMPLE RETURN

$$R_t = \frac{P_{t+1} - P_t}{P_t} = \frac{P_{t+1}}{P_t} - 1$$

When dividends are included, the total return becomes:

TOTAL RETURN WITH DIVIDENDS

$$R_t = \frac{P_{t+1} - P_t + D_t}{P_t}$$

where P_t is the price at time t , P_{t+1} is the price at time $t + 1$, and D_t represents dividends paid during the period.

IV. Arithmetic vs. Geometric Rates of Return

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Two different averaging methods exist for summarizing returns over multiple periods:

ARITHMETIC MEAN RETURN

$$\bar{R}_A = \frac{1}{T} \sum_{t=1}^T R_t$$

GEOMETRIC MEAN RETURN

$$\bar{R}_G = \left[\prod_{t=1}^T (1 + R_t) \right]^{1/T} - 1$$

Consider an illustrative example: $P_0 = \$100$, $R_1 = -50\%$, $R_2 = +100\%$. After the first period, wealth falls to 50; *after the second, it returns to 100*. The arithmetic average return is $\frac{-50\% + 100\%}{2} = 25\%$, but the geometric average is $\sqrt{(0.50)(2.00)} - 1 = 0\%$.

Key Concept: Arithmetic vs. Geometric Returns

The **geometric return** better reflects the actual investment experience (compound growth), while the **arithmetic return** suits single-period statistical models. The geometric mean is always less than or equal to the arithmetic mean, with the gap increasing as volatility rises.

V. Capital Market History

The 1980s

The 1980s represented exceptional returns for stock investors, with S&P 500 performance showing substantial growth.

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The 1930s

This decade experienced severe market crashes globally:

U.S. stock investors experienced a zero percent return for the eleven-year period from 12/1929 to 12/1939.

Long-Term Performance (1926–1995)

Over 68 years, a dollar invested in the S&P 500 grew to approximately 800, *while corporate bonds grew to only 40*. The stock line displays far greater volatility than the bond line, but the terminal wealth difference is dramatic.

Data note: The original text used Ibbotson Associates data through March 1995. Modern Ibbotson/Morningstar data extending through recent years shows the same fundamental patterns — stocks dramatically outperform bonds over long horizons, but with substantially greater year-to-year volatility.

Growth of \$1 Invested in 1926

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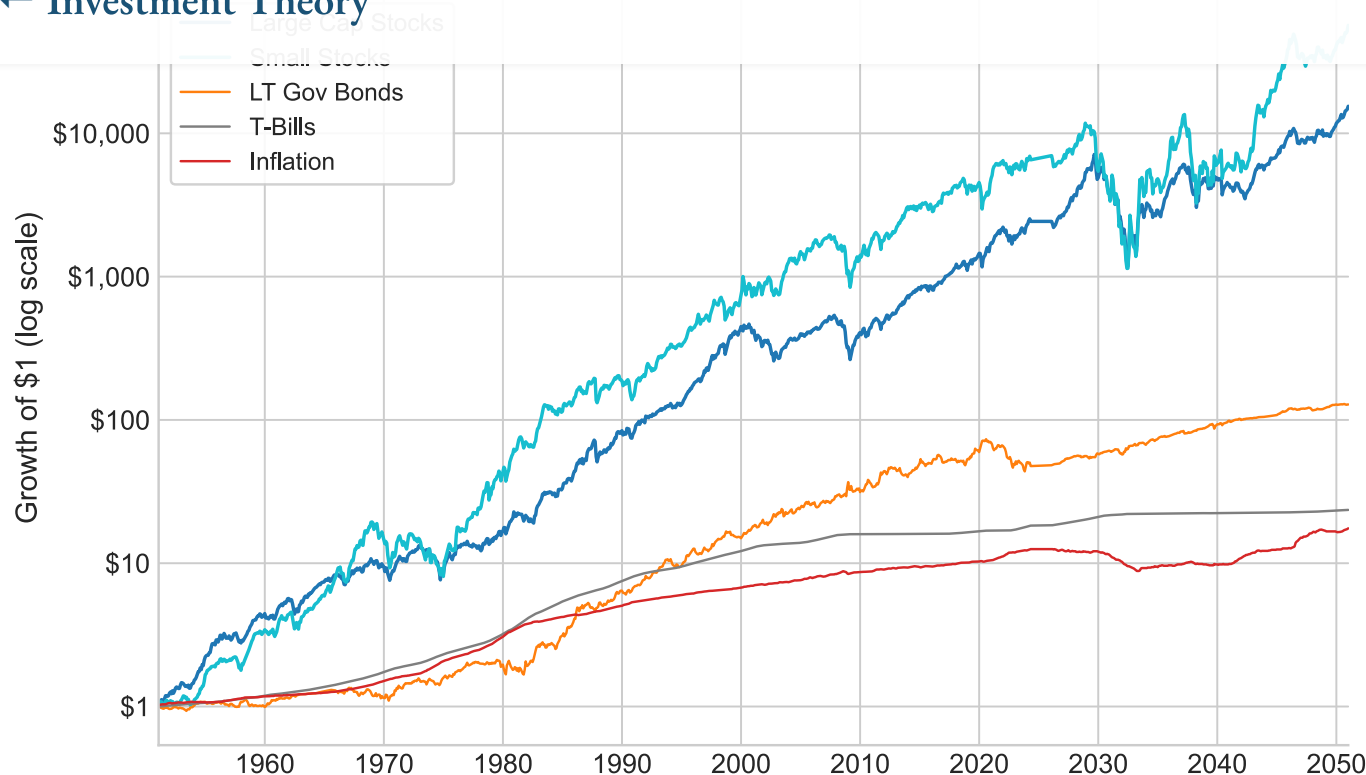


Figure 1.1. Growth of \$1 invested in U.S. stocks, bonds, and Treasury bills (1926–2024, log scale). Computed from SBBi data. Stocks dramatically outperform bonds over long horizons, but with substantially greater year-to-year volatility.

VI. The Risk Premium

The return differential between stocks and bonds is attributed to risk differences.

"Volatility" describes the shakier performance of stocks compared to bonds. Investors are **risk-averse** — they prefer less risk when other factors remain equal. This motivates the concept of the **risk premium**: the additional return demanded for holding risky securities versus riskless alternatives like Treasury Bills.

EQUITY PREMIUM

$$\text{Equity Premium} = \bar{R}_{\text{stocks}} - \bar{R}_{\text{T-bills}}$$

From 1926–1995, the equity premium was approximately 9% (arithmetic) or 6% (geometric) annually.

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Summary Statistics (1926–March 1995)

INVESTMENT	GEOMETRIC MEAN	ARITHMETIC MEAN	STD. DEV.	HIGH RETURN	LOW RETURN
S&P 500 Total Return	10.30%	12.45%	22.28%	42.56%	−29.73%
U.S. Small Stocks TR	12.28%	17.28%	35.94%	73.46%	−36.74%
U.S. LT Govt. Bonds TR	4.91%	5.21%	8.00%	15.23%	−8.41%
U.S. LT Corp. Bonds TR	5.49%	5.73%	7.16%	13.76%	−8.90%
U.S. 30-day T-Bills	3.70%	3.70%	0.96%	1.35%	−0.06%

Source: Ibbotson Associates

VII. Standard Deviation as a Measure of Risk

Standard deviation quantifies volatility mathematically as the square root of the variance. It calculates the average spread of observations around the mean:

STANDARD DEVIATION

$$\sigma = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2}$$

For S&P 500 returns (approximately $\sigma = 22.28\%$), assuming a normal distribution,

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approximately two-thirds of observations should fall within one standard deviation of

the mean:

$$\bar{R} \pm \sigma \approx 12.45\% \pm 22.28\% = [-9.83\%, 34.73\%]$$

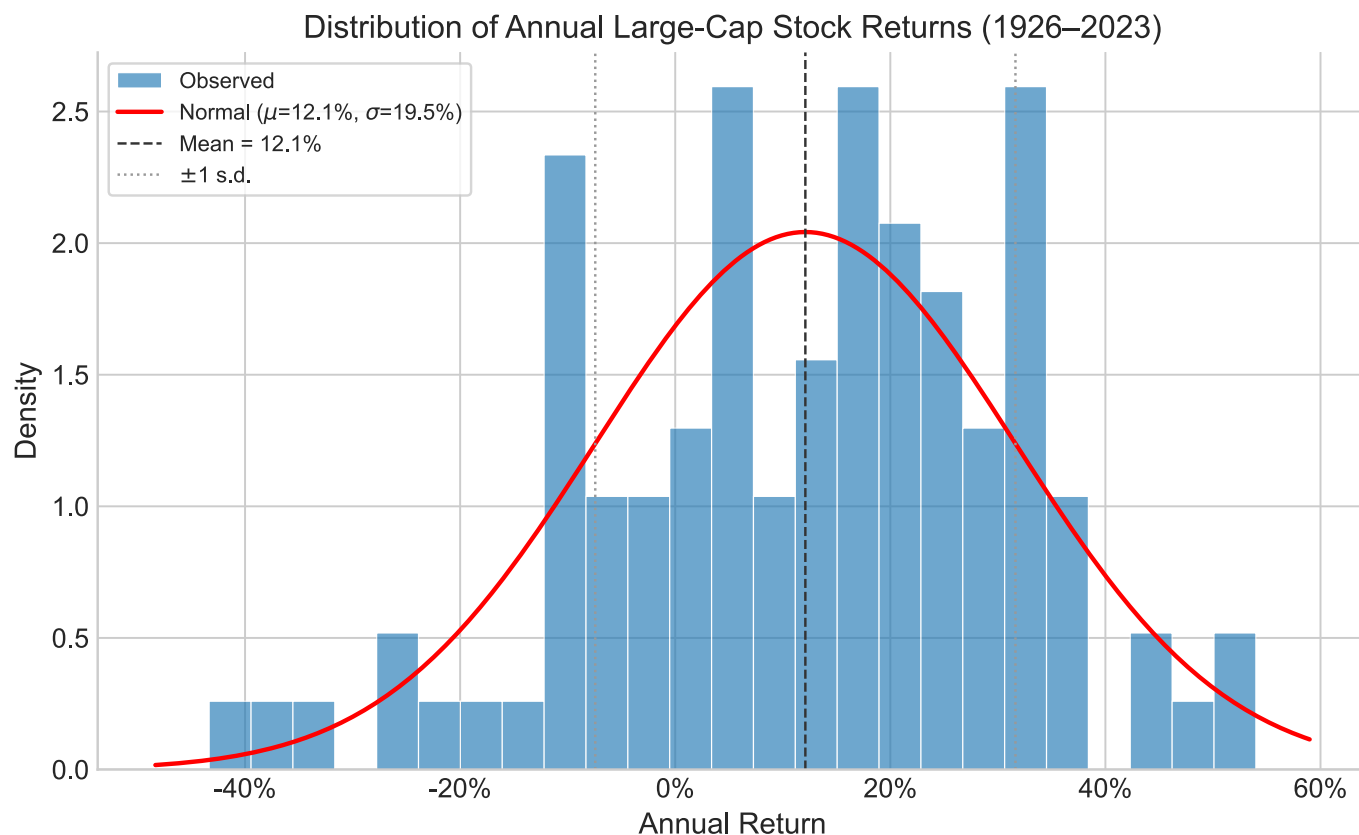


Figure 1.2. *Distribution of S&P 500 annual returns, computed from SBBI data. The histogram shows the ± 1 standard deviation range containing roughly 68% of observations. Actual return distributions exhibit "fatter tails" than the normal distribution predicts.*

Limitations of Standard Deviation

The chapter acknowledges several limitations:

- Standard deviation equally weights high and low returns

- It heavily weights extreme observations
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- It ignores distribution shape (skewness and kurtosis)

However, the benefits include providing a single comparable risk measure enabling portfolio analysis decisions.

There is evidence that stock returns may follow "stable" distributions with undefined variance rather than normal distributions, a hypothesis advanced by Benoit Mandelbrot. Return distributions show "fatter tails" than log-normal predictions suggest — extreme events (both positive and negative) occur more frequently than a normal model predicts.

Key Concept: Standard Deviation as Risk

Standard deviation provides a single, comparable measure of risk that enables systematic portfolio analysis. Despite its limitations — especially its assumption of symmetric, normally distributed returns — it remains the foundational risk measure in modern portfolio theory.

Interactive Tool: Explore these concepts interactively using the [Rice Business Investments Compendium](#), created by Kerry Back and Kevin Crotty at the Jones Graduate School of Business, Rice University.

Acknowledgement: The interactive tools referenced in this textbook are from the [Rice Business Investments Compendium](#), created and maintained by Kerry Back and Kevin Crotty at the Jones Graduate School of Business, Rice University. We gratefully acknowledge their contribution to investment education.

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